*F18CF – Linear Algebra*

# Tutorial Sheet 4 – Solutions

1. (a) The set $S_1$ does not contain the zero vector $\vec{0} = (0,0,0)^T$ since an element of $S_1$ always its second coordinate equal to 1. Since a subspace always contains $\vec{0}$, it follows that $S_1$ is not a subspace of $\mathbb{R}^3$.

   (b) The set $S_2$ is the set of vectors of the form

   $$\begin{pmatrix} a \\ b \\ a \end{pmatrix} = a \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + b \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

   for some $a, b \in \mathbb{R}$. In other words,

   $$S_2 = \text{span}\left( \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right).$$

   Since we know a span of vectors is always a subspace, $S_2$ is a subspace of $\mathbb{R}^3$.

   (c) $S_3$ is the set of solutions of a linear homogeneous equation. From the course, it follows that $S_3$ is a subspace of $\mathbb{R}^3$.

   (d) Let $\vec{u} = (1,0,0)^T$ and $\vec{v} = (0,1,0)^T$. We have $\vec{u}, \vec{v} \in S_4$ since $1^2 + 0^2 + 0^2 = 0^2 + 1^2 + 0^2 = 1$. However, $\vec{u} + \vec{v} = (1,0,0)^T + (0,1,0)^T = (1,1,0)^T$ is not in $S_4$ since $1^2 + 1^2 + 0^2 = 2 > 1$. Since $S_4$ is not stable under sums of vectors, it follows that $S_4$ is not a subspace of $\mathbb{R}^3$.

2. (a) We solve this system using Gaussian elimination:

   $$\left(\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 2 & -1 & 1 & 0 \\ 0 & 3 & 3 & 0 \end{array}\right) \overset{R_2 \to R_2 - 2R_1}{\rightsquigarrow} \left(\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 0 & -3 & -3 & 0 \\ 0 & 3 & 3 & 0 \end{array}\right) \overset{R_3 \to R_3 + R_2}{\rightsquigarrow} \left(\begin{array}{ccc|c} 1 & 1 & 2 & 0 \\ 0 & -3 & -3 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right)$$

   There is one free variable $z$, so we set $z = \alpha$, $\alpha \in \mathbb{R}$, and by substitution we get $y = -\alpha$, $x = -\alpha$.
   Hence the general solution is of the form

   $$\begin{pmatrix} -\alpha \\ -\alpha \\ \alpha \end{pmatrix} = \alpha \begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}.$$

   Therefore, the space of solution has dimension 1 and a basis is given by the vector $(-1,-1,1)^T$.

(b) We solve this system using Gaussian elimination:

$$\left(\begin{array}{cccc|c} -1 & 0 & 1 & 2 & 0 \\ -1 & 1 & 0 & -1 & 0 \\ 0 & -1 & 1 & 3 & 0 \\ 1 & -2 & 1 & 4 & 0 \end{array}\right) \overset{\substack{R_2\to R_2-R_1 \\ R_4\to R_4+R_1}}{\rightsquigarrow} \left(\begin{array}{cccc|c} -1 & 0 & 1 & 2 & 0 \\ 0 & 1 & -1 & -3 & 0 \\ 0 & -1 & 1 & 3 & 0 \\ 0 & -2 & 2 & 6 & 0 \end{array}\right) \overset{\substack{R_3\to R_3+R_2 \\ R_4\to R_4+2R_2}}{\rightsquigarrow}$$

$$\left(\begin{array}{cccc|c} -1 & 0 & 1 & 2 & 0 \\ 0 & 1 & -1 & -3 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right)$$

We thus have two free variables $z, u$, so we set $u = \alpha$, $z = \beta$, $\alpha, \beta \in \mathbb{R}$, and we get by substitution $y = 3\alpha + \beta$ and $x = 2\alpha + \beta$. The general solution is thus of the form

$$\begin{pmatrix} 2\alpha+\beta \\ 3\alpha+\beta \\ \beta \\ \alpha \end{pmatrix} = \alpha \begin{pmatrix} 2 \\ 3 \\ 0 \\ 1 \end{pmatrix} + \beta \begin{pmatrix} 1 \\ 1 \\ 1 \\ 0 \end{pmatrix}.$$

Thus, the space of solutions has dimension 2, and a basis of it is given by the vectors $(2,3,0,1)^T,\ (1,1,1,0)^T$.

3. (a) We compute a basis by performing Gaussian elimination on the associated matrix:

$$\begin{pmatrix} 1 & 4 & 3 \\ 3 & 0 & 1 \\ 4 & 1 & 2 \end{pmatrix} \overset{\substack{R_2\to R_2-3R_1 \\ R_3\to R_3-4R_1}}{\rightsquigarrow} \begin{pmatrix} 1 & 3 & 4 \\ 0 & -12 & -8 \\ 0 & -15 & -10 \end{pmatrix} \overset{R_3\to 4R_3-5R_2}{\rightsquigarrow} \begin{pmatrix} 1 & 3 & 4 \\ 0 & -12 & -8 \\ 0 & 0 & 0 \end{pmatrix}$$

There are two pivot variables corresponding to the first two columns, so span $(\vec{u}_1, \vec{u}_2, \vec{u}_3)$ has dimension 2 and a basis for it is given by $\vec{u}_1, \vec{u}_2$.

(b) We compute a basis by performing Gaussian elimination on the associated matrix:

$$\begin{pmatrix} 2 & 4 & 1 & 1 \\ 1 & 8 & -3 & 10 \\ 0 & -4 & 2 & -6 \\ -1 & -3 & 0 & -2 \end{pmatrix} \overset{\substack{R_2\to 2R_2-R_1 \\ R_4\to 2R_4+R_1}}{\rightsquigarrow} \begin{pmatrix} 1 & -3 & 2 & 0 \\ 0 & 12 & -7 & 19 \\ 0 & -4 & 2 & 6 \\ 0 & -2 & 1 & -3 \end{pmatrix} \overset{\substack{R_3\to 3R_3+R_2 \\ R_4\to 6R_4+R_2}}{\rightsquigarrow}$$

$$\begin{pmatrix} 1 & -3 & 2 & 0 \\ 0 & 12 & -7 & 19 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & -1 & 1 \end{pmatrix} \overset{R_4\to R_4-R_3}{\rightsquigarrow} \begin{pmatrix} 1 & -3 & 2 & 0 \\ 0 & 12 & -7 & 19 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

There are three pivot variables corresponding to the first three columns, so span $(\vec{u}_1, \vec{u}_2, \vec{u}_3, \vec{u}_4)$ has dimension 3 and a basis for it is given by $\vec{u}_1, \vec{u}_2, \vec{u}_3$.

4. (a) The zero function clearly belongs to $S_1$. Suppose $P, Q \in S_1$ and $\alpha \in \mathbb{R}$.
Then $(P+Q)(1) = P(1) + Q(1) = 0 + 0 = 0$ and so $P + Q \in S_1$.

Also $(\alpha P)(1) = \alpha P(1) = \alpha \times 0 = 0$ so $\alpha P \in S_1$.
Hence $S_1$ is a subspace of $\mathcal{P}(\mathbb{R})$.

(b) The zero function is not in $S_2$, so $S_2$ is not a subspace of $\mathcal{P}(\mathbb{R})$.

(c) The zero function clearly belongs to $S_3$. Suppose $P, Q \in S_3$ and $\alpha \in \mathbb{R}$.
As was shown in the the first year calculus course, we have $(P+Q)'(0) = P'(0)+Q'(0) = 0+0 = 0$ and so $P+Q \in S_3$. Also $(\alpha P)'(0) = \alpha P'(0) = \alpha \times 0 = 0$ so $\alpha P \in S_3$.
Hence $S$ is a subspace of $\mathcal{P}(\mathbb{R})$.

5. (a) It is a subspace, as the solution space of a homogeneous system of (one) linear equation, by a result of the course.

(b) The matrix associated to this system is

$$\begin{pmatrix} a_1 & a_2 & \dots & a_n \end{pmatrix},$$

which is in reduced form. In particular, since the $a_i$ are not all zero, there exists exactly one pivot variable and $n-1$ free variables, hence the hyperplane has dimension $n-1$.

(c) The matrix associated to this system is

$$\begin{pmatrix} 1 & 2 & 3 & 4 \end{pmatrix},$$

which is in reduced form. There are three free variables $x_2, x_3, x_4$. Setting $x_4 = \alpha, x_3 = \beta, x_2 = \gamma$, we get by substitution $x_1 = -4\alpha - 3\beta - 2\gamma$. In particular, the general solution is of the form

$$\begin{pmatrix} -4\alpha - 3\beta - 2\gamma \\ \gamma \\ \beta \\ \alpha \end{pmatrix} = \alpha \begin{pmatrix} -4 \\ 0 \\ 0 \\ 1 \end{pmatrix} + \beta \begin{pmatrix} -3 \\ 0 \\ 1 \\ 0 \end{pmatrix} + \gamma \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix}.$$

Thus, a basis for that hyperplane is $(-4,0,0,1)^T, (-3,0,1,0)^T, (-2,1,0,0)^T$.

6. (a) We check the properties of subspaces. The null vector $\vec{0}$ belongs to $V$ and $W$ (as they are subspaces), hence it belongs to the intersection $V \cap W$. Let $\vec{v}_1, \vec{v}_2 \in V \cap W$ and $\lambda \in \mathbb{R}$. Then in particular, we have $\vec{v}_1, \vec{v}_2 \in V$ and since $V$ is a subspace of $\mathbb{R}^n$, we have $\vec{v}_1 + \vec{v}_2 \in V$ and $\lambda \vec{v}_1 \in V$. Similarly, we have $\vec{v}_1, \vec{v}_2 \in W$ and since $W$ is a subspace of $\mathbb{R}^n$, we have $\vec{v}_1 + \vec{v}_2 \in W$ and $\lambda \vec{v}_1 \in W$. We thus get $\vec{v}_1 + \vec{v}_2 \in V \cap W$ and $\lambda \vec{v}_1 \in V \cap W$, hence $V \cap W$ is a subspace of $\mathbb{R}^n$.

(b) Let $\vec{v}, \vec{w} \in \mathbb{R}^n$ be two linearly independent vectors, and take $V$ and $W$ two be the one-dimensional subspace spanned by $\vec{v}$ and $\vec{w}$ respectively. Then $V \cap W = \{\vec{0}\}$.

(c) The union $V \cup W$ is not always a subspace of $\mathbb{R}^n$. For instance, let $\vec{v}, \vec{w} \in \mathbb{R}^n$ be two linearly independent vectors, and take $V$ and $W$ two be the one-dimensional subspace spanned by $\vec{v}$ and $\vec{w}$ respectively. Then $\vec{v}, \vec{w} \in V \cup W$ by construction, but their sum $\vec{v} + \vec{w}$ is not in $V \cup W$: If $\vec{v} + \vec{w}$ was in $V \cup W$, or in other words was a multiple of $\vec{v}$ or of $\vec{w}$, then we would obtain a non-trivial relation between $vv$ and $\vec{w}$, contradicting the fact that their are linearly independent. Thus, Then $\vec{v}, \vec{w} \in V \cup W$ but $\vec{v} + \vec{w} \notin V \cup W$, hence $V \cup W$ is not a subspace of $\mathbb{R}^n$.

7. (a) The zero vector $\vec{z} := (0, 0, \dots, )$ belongs to $V$ as the equation $0 = 4 \times 0 - 4 \times 0$ is indeed satisfied for every $n$.

Let $u = (u_n)_n$ and $v = (v_n)_n$ be two sequences in $V$, and let $\alpha \in \mathbb{R}$. We have for every $n \geq 0$:

$$\begin{aligned}(u+v)_{n+2} = u_{n+2} + v_{n+2} &= 4u_{n+1} - 4u_n + 4v_{n+1} - 4v_n \\ &= 4(u_{n+1} + v_{n+1}) - 4(u_n + v_n) = 4(u+v)_{n+1} - 4(u+v)_n,\end{aligned}$$

so $u + v \in V$. Moreover, we have for $n \geq 0$:

$$(\alpha u)_{n+2} = \alpha u_{n+2} = \alpha(4u_{n+1} - 4u_n) = 4(\alpha u)_{n+1} - 4(\alpha u)_n,$$

so $\alpha u \in V$. Thus, $V$ is a subspace of $\mathbb{R}^{\mathbb{N}}$.

(b) We have $2^{n+2} = 4 \times 2^n$ and $4 \times 2^{n+1} - 4 \times 2^n = 8 \times 2^n - 4 \times 2^n = 4 \times 2^n$. Thus we have $2^{n+2} = 4 \times 2^{n+1} - 4 \times 2^n$ for every $n \geq 0$, hence the sequence $2^n$ is in $V$.

Similarly, we have $(n+2)2^{n+2} = 4n \times 2^n + 8 \times 2^n$ and $4(n+1) \times 2^{n+1} - 4n \times 2^n = 8n \times 2^n + 8 \times 2^n - 4n \times 2^n = 4n \times 2^n + 8 \times 2^n$. Thus we have $(n+2)2^{n+2} = 4(n+1) \times 2^{n+1} - 4n2^n$ for every $n \geq 0$, hence the sequence $n2^n$ is in $V$.

(c) We consider the sequences $v := (v_n)$ and $w := (w_n)$. Let $\alpha, \beta \in \mathbb{R}$ such that $\alpha v + \beta v = \vec{0}$. This can be rewritten as

$$\alpha \begin{pmatrix} 1 \\ 2 \\ 4 \\ 8 \\ \vdots \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 2 \\ 8 \\ 24 \\ \vdots \end{pmatrix} = \begin{pmatrix} \alpha \\ 2\alpha + 2\beta \\ 4\alpha + 8\beta \\ 8\alpha + 24\beta \\ \vdots \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ \vdots \end{pmatrix}.$$

Two sequences are equal if and only if their terms agree. By taking the first two coordinates, we get

$$\begin{cases} \alpha & = & 0 \\ 2\alpha + 2\beta & = & 0 \end{cases}$$

which yields, $\alpha = \beta = 0$. Thus, $v$ and $w$ are linearly independent.

8. (a) The zero sequence is clearly periodic (of period 1 for instance), hence is in $S_1$. Note that if a sequence is $T$-periodic, it is also $kT$-periodic for every integer $k \geq 1$. Let $(u_n)$, $(v_n)$ be two periodic sequences in $S_1$, with period $T$ and $S$ respectively. Since $(u_n)$ is $T$-periodic, it is also $TS$-periodic. Moreover, since $(v_n)$ is $S$-periodic, it is also $ST = TS$-periodic. Let us show that the $(u_n) + (v_n)$ is $TS$-periodic. For every integer $n \geq 0$, we have

$$u_{n+TS} + v_{n+TS} = u_n + v_n,$$

since both sequences are $TS$-periodic, hence the sequence $(u_n) + (v_n)$ is $TS$-periodic, hence is in $S_1$.

Moreover, for every $\lambda \in \mathbb{R}$, we have

$$\lambda u_{n+T} = \lambda u_n$$

since $(u_n)$ is $T$-periodic, hence the sequence $\lambda(u_n)$ is $T$-periodic, hence in $S_1$.

Thus, $S_1$ is a subspace of $\mathbb{R}^{\mathbb{N}}$.

(b) The zero function is clearly $2\pi$-periodic, hence in $S_2$. Let $f, g \in S_2$ and let $\lambda \in \mathbb{R}$. For every $x \in \mathbb{R}$, we have

$$(f+g)(x+2\pi) = f(x+2\pi) + g(x+2\pi) = f(x) + g(x) = (f+g)(x),$$

hence $f + g \in S_2$ We also have for every $x \in \mathbb{R}$:

$$(\lambda f)(x+2\pi) = \lambda \times f(x+2\pi) = \lambda \times f(x) = (\lambda f)(x),$$

hence $\lambda f \in S_2$.

Thus, $S_2$ is a subspace of $\mathcal{F}(\mathbb{R})$.

(c) Consider the functions $f$, $g \in \mathcal{F}(\mathbb{R})$ defined by

$$f(x) = \cos(x), \quad g(x) = \cos(x\sqrt{2}) \quad \text{for all } x \in \mathbb{R}.$$

Both $f$ and $g$ are periodic (of period $2\pi$ and $\sqrt{2}\pi$ respectively), hence are in $S_3$. Let us show by contradiction that $f + g$ is not in $S_3$. If $f + g$ was periodic, there would exist a real $T > 0$ such that for every $x \in \mathbb{R}$,

$$\cos(x+T) + \cos(\sqrt{2}(x+T)) = \cos(x) + \cos(x\sqrt{2}).$$

In particular, for $x = 0$ we get $\cos(T) + \cos(T\sqrt{2}) = 2$. But since $|\cos(x)| \leq 1$ for all $x \in \mathbb{R}$, this means that $\cos(T) = \cos(T\sqrt{2}) = 1$. Thus, there exist integers $k, \ell > 0$ such that $T = 2k\pi$ and $T\sqrt{2} = 2\ell\pi$. We thus get

$$\sqrt{2} = \frac{T\sqrt{2}}{T} = \frac{2k\pi}{2\ell\pi} = \frac{k}{\ell},$$

which is absurd since $\sqrt{2}$ is an irrational number. Thus, $f + g$ is not periodic.

Since $S_3$ is not stable under sums, it is not a subspace of $\mathcal{F}(\mathbb{R})$.